# Spherical Rate–Distortion-Guided Compression Panoramic 3D Gaussian Splatting

***Abstract*—Panoramic 3D Gaussian Splatting (3DGS) enables efficient 360° novel-view synthesis but incurs substantial transmission and storage challenges. Specifically, conventional Spherical Harmonic (SH) capacity allocation based on coefficient energy fails to reflect rendering contributions, while the nonuniform sampling of Equi-Rectangular Projection (ERP) causes a mismatch between pixel-domain and spherical distortion. To address these issues, we develop a panoramic 3DGS compression framework. Specifically, spherical-area-weighted rendering gradients and conditional rate estimates guide reversible SH-band exchanges to redistribute appearance capacity under a fixed bit budget. Candidate exchanges are validated through re-rendering and actual bitstream-size checks, with all optimization performed during offline encoding. Experiments across 43 scenes demonstrate improved rate–distortion performance over energy-based allocation and additional benefits from our method.**

***Index Terms*—Panoramic rendering, 3D Gaussian splatting, compression, rate–distortion optimization.**

## I. Introduction

3D Gaussian Splatting (3DGS) achieves high-quality, real-time novel view synthesis through explicit Gaussian primitives [1]. Traditional 3DGS primarily relies on perspective images with limited fields of view for scene reconstruction and rendering. In contrast, panoramic images capture a wider scene coverage in a single acquisition, improving data collection efficiency and providing more comprehensive environmental information [3]. Panoramic 3DGS supports a 360° omnidirectional field of view, making it particularly suitable for virtual reality and immersive scene navigation [17], [18]. OmniGS [3] further enables native equirectangular projection (ERP) rendering without decomposing panoramic images into multiple perspective views. However, 3DGS requires storing a large number of Gaussian primitives and their associated attributes, and the increased data volume of panoramic scenes poses significant challenges for storage and transmission [4].

Existing studies have improved the efficiency of 3DGS from two perspectives: rendering and compression [2], [8]. For panoramic rendering, OmniGS [3] enables efficient ERP rendering. For compression, LightGaussian [4], MesonGS [5], HAC++ [6], and PCGS [7] reduce storage overhead through techniques such as Gaussian pruning, attribute transformation, and entropy coding. However, existing general-purpose 3DGS compression methods are primarily designed for perspective views or general scene representations, without adequately considering the rate–distortion characteristics of panoramic rendering. Specifically, SH coefficient energy does not necessarily reflect its contribution to the final rendering quality. Allocating representation capacity solely based on static attributes may therefore fail to prioritize Gaussian primitives that significantly affect reconstruction quality [8]. Furthermore, ERP images exhibit nonuniform sampling on the sphere, with oversampling at high latitudes. Conventional pixel-uniform distortion metrics cannot accurately measure reconstruction errors across different spherical regions [16]. Therefore, panoramic 3DGS compression needs to jointly consider the actual rendering contributions of Gaussian attributes and the characteristics of spherical distortion.

To address these challenges, this paper proposes a spherical rate–distortion-guided compression method for panoramic 3DGS. The central idea is to evaluate coding decisions by their effects on panoramic reconstruction quality, so that limited storage is allocated according to its benefit on the sphere. The resulting representation supports independent decoding and native ERP rendering, with all compression optimization confined to offline encoding.

Our contributions are threefold: (1)To the best of our knowledge, this is the first work that presents the dedicated compression framework for panoramic 3D Gaussian Splatting, which realizes joint optimization of 3DGS compression and 360° panoramic rendering.; (2)A rate-distortion guided spherical harmonic (SH) capacity re-allocation strategy is proposed for panoramic scenarios, where reversible SH-band swaps are evaluated within an offline closed-loop pipeline to redistribute SH capacity under given bit-budget constraints ; and (3) to address the pixel-to-spherical domain distortion mismatch originating from the non-uniform sampling nature of equirectangular projection (ERP), spherical-area weighting is incorporated into the rate-distortion objective, where reconstruction errors are weighted by their corresponding actual spherical surface footprint instead of raw pixels.

The source code is publicly available to support reproducibility.[1]

## II. Proposed Method

### A. Motivation for Spherical Rate–Distortion Optimization

Compressing 3D Gaussian representations for panoramic scenes requires reducing storage costs while preserving omnidirectional rendering quality. However, applying existing 3DGS compression methods to panoramic scenes does not necessarily yield compact representations with consistent visual quality. As shown in Fig. 3, under the evaluated compression configurations, HAC++ and PCGS exhibit noticeable texture blurring and indistinct boundaries in the rocky regions of the coastal scene, while LightGaussian produces pronounced appearance artifacts. In the bedroom scene, curtain folds and the contours of objects on the desk also become blurred in the reconstructions of several competing methods. These degradations indicate that existing compression strategies still

[1] https://github.com/PPMYfly/SCP-GS

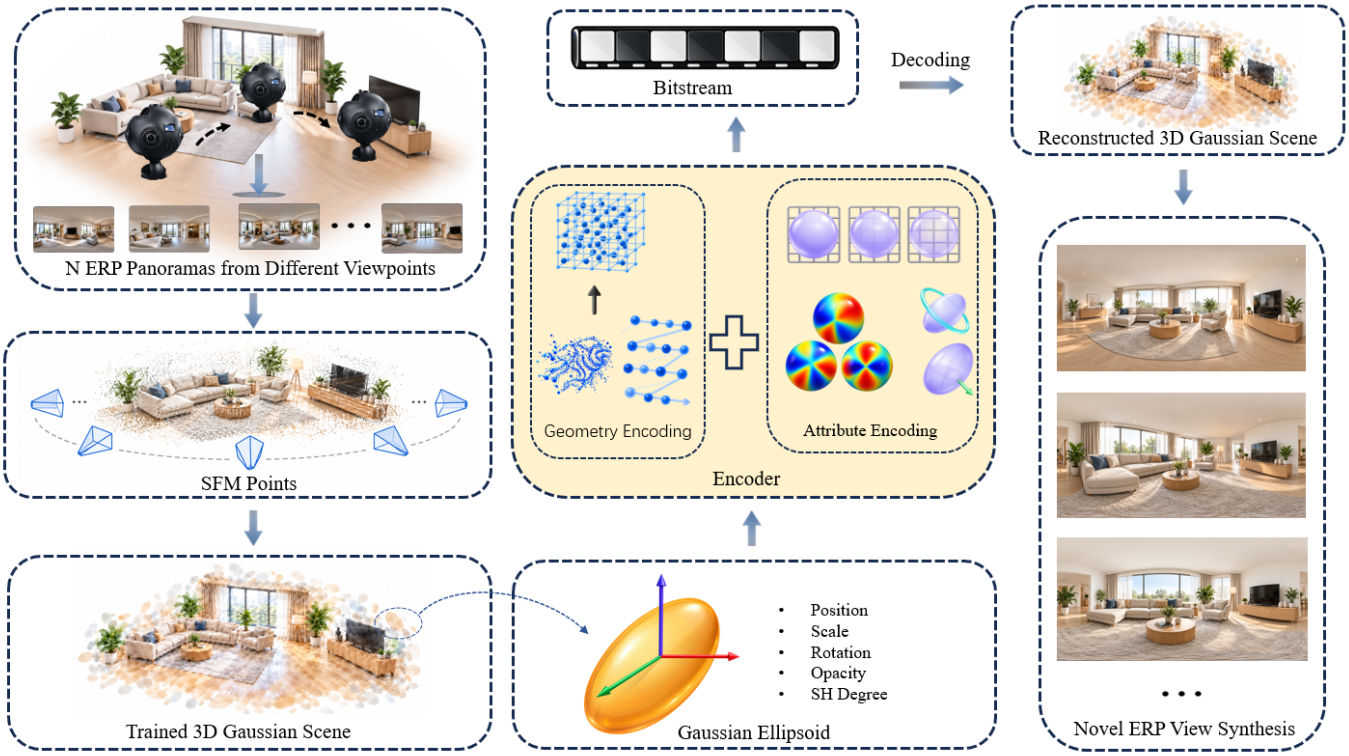



Fig. 1. Overview of panoramic 3D Gaussian compression. Scene reconstruction initializes the source Gaussian field; geometry and variable-degree SH attributes are encoded into a self-contained bitstream for independent decoding and novel-view rendering. The offline feedback loop reallocates SH capacity using spherical distortion and actual-byte validation.

struggle to balance representation size and rendering fidelity in the evaluated panoramic scenes.

These observations suggest that the effects of compression errors should be evaluated in the final rendering space. Gaussian parameter perturbations of similar magnitudes may affect panoramic images differently depending on visibility, projected extent, and view-dependent appearance. Therefore, allocating appearance capacity solely according to static attributes, such as SH coefficient energy, may not accurately reflect the contribution of each Gaussian to rendering quality. Moreover, ERP pixels correspond to different spherical areas, so uniformly weighted pixel errors may not accurately measure spherical distortion across regions. Without sufficient consideration of panoramic rendering feedback and spherical observation characteristics, compression decisions may fail to allocate the limited coding budget to appearance information that matters most to the final image.

These observations motivate us to couple panoramic representation compression with final rendering quality and allocate the limited coding budget under a spherical rate–distortion objective. Specifically, compression should exploit statistical redundancy among Gaussian parameters while accounting for the rendering contributions of different regions and appearance components, thereby balancing appearance capacity and coding cost under an actual bitstream budget. Our goal is thus to obtain a compact panoramic Gaussian representation that reduces storage costs while preserving important structural details and view-dependent appearance.

### *B. Panoramic Gaussian Compression with Spherical Rendering Feedback*

As illustrated in Fig. 1, the proposed framework takes ERP panoramas as input and reconstructs a trained 3D Gaussian scene through SfM initialization and iterative optimization. The encoder compresses geometry using quantization, Morton ordering, and G-PCC, while appearance attributes are encoded with variable SH degrees . To improve rate–distortion efficiency, spherical rendering feedback guides SH capacity reallocation through gradient-based candidate scoring and ERP re-rendering validation under the actual bitstream budget. The resulting self-contained bitstream is independently decoded into a 3D Gaussian scene for native ERP novel-view rendering.

As illustrated in Fig. 2, we optimize the SH degrees of Gaussians to minimize spherical rendering distortion under a bitrate budget constraint. This optimization problem is formulated as

$$\min_{\boldsymbol{\ell}} D_{\mathrm{sph}}\left(\mathcal{R}_{\mathrm{ERP}}(\widehat{G}(\boldsymbol{\ell})), I^{\mathrm{ref}}\right), \quad \text{s.t. } B(\boldsymbol{\ell}) \le B_{\max}, \quad (1)$$

where $\boldsymbol{\ell} = (\ell_1, \ldots, \ell_N)$ denotes the SH degree configuration of all Gaussians, with $\ell_i \in \{0, \ldots, L\}$, where $L \le 3$ is the maximum source SH degree shared by all Gaussians. Here, $\widehat{G}(\boldsymbol{\ell})$ denotes the corresponding reconstructed Gaussians, $\mathcal{R}_{\mathrm{ERP}}$ denotes panoramic rendering, and $I^{\mathrm{ref}}$ represents the reference rendering of the trained source Gaussians. Furthermore, $D_{\mathrm{sph}}$ denotes the spherical-area-weighted rendering distortion, $B(\boldsymbol{\ell})$ is the bitrate after 3DGS compression, and

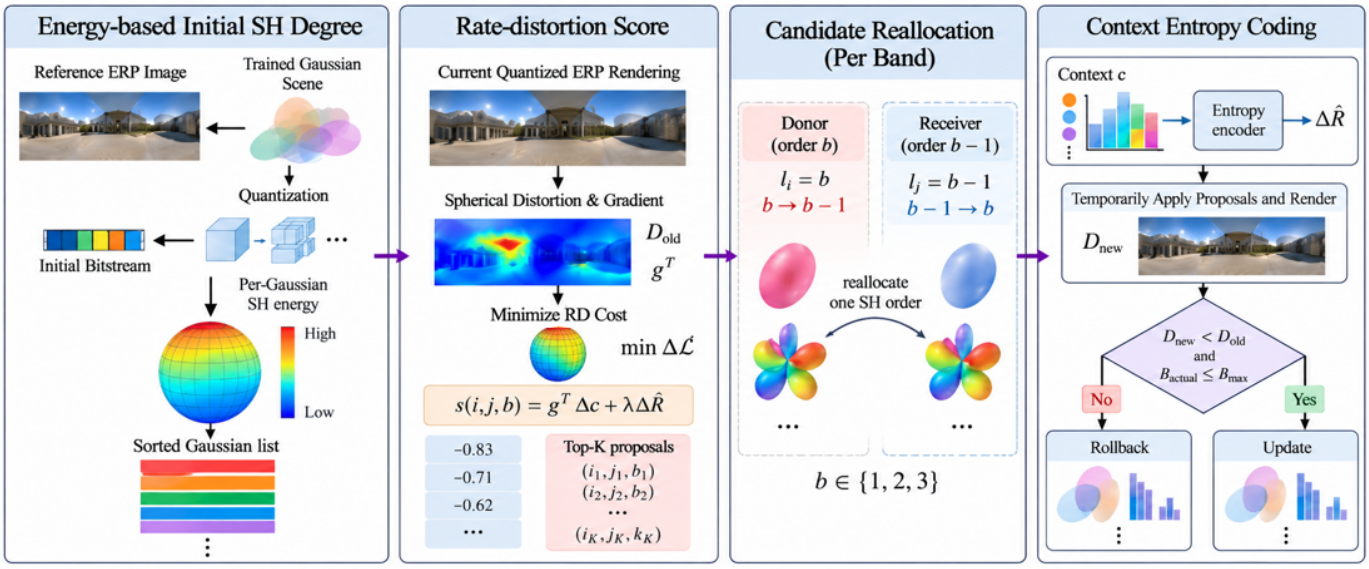



Fig. 2. Spherical rendering feedback-guided SH reallocation. Starting from energy-based SH degrees, spherical distortion gradients and conditional rate estimates score per-band donor–receiver exchanges. Proposed exchanges are validated by ERP re-rendering and actual bitstream size; rejected proposals are rolled back.

$B_{\max}$ is the bitrate budget. Here, bitrate is defined as the total number of coded bits per scene, including all information required for independent decoding.

Directly solving this optimization problem is time consuming because SH degrees are discrete variables, while distortion and actual rate evaluation require panoramic rendering and complete encoding, respectively. Therefore, as illustrated in the proposed workflow, we first initialize the SH degrees based on coefficient energy, then identify promising reallocation candidates using spherical rendering gradients and conditional entropy estimates. The selected candidates are subsequently validated through actual rendering and complete encoding.

*1) Energy-Based Initial SH Degree Allocation:* We first compute the SH band energy of each Gaussian. Let $E_{i,b} = \|\mathbf{c}_{i,b}\|_2^2$ denote the source coefficient energy of SH band $b$ for Gaussian $i$. The energy of discarded bands is used as a distortion proxy, while the estimated bits required for retained SH coefficients and degree signaling serve as a rate proxy. The same allocation rule is applied to all Gaussians, and the initial SH degree is determined by

$$\ell_i^{(0)} = \underset{k\in\{0,\dots,L\}}{\arg\min} \left[ \sum_{b=k+1}^{L} E_{i,b} + \mu\big(pS(k) + h_i(k)\big) \right], \quad (2)$$

where $S(k) \in \{0, 9, 24, 45\}$ denotes the total number of non-DC SH scalar coefficients retained up to degree $k$, $p$ is a common per-coefficient bit-cost estimate, and $h_i(k)$ represents the estimated bit cost of signaling the selected SH degree. The scene-level energy normalization factor is absorbed into $\mu$, which controls the trade-off between distortion and rate. Its value is determined by binary search to satisfy a predefined initial SH capacity target.

After the initial degree allocation, the Gaussians are fully encoded, and the resulting actual bitrate is used to define the budget $B_{\max}$ for subsequent SH capacity reallocation.

*2) SH Capacity Reallocation Guided by Spherical Rendering Feedback: Spherical Rate?Distortion Scoring.* At each iteration, the current reconstructed Gaussians are first rendered using the panoramic renderer, with the renderings of the trained source Gaussian scene from the same viewpoints serving as references. To account for the nonuniform spherical areas represented by pixels at different latitudes, in the ERP case, we measure rendering distortion using spherical-area-weighted mean squared error (WS-MSE):

$$D_{\text{sph}} = \frac{1}{|\mathcal{V}|} \sum_{v\in\mathcal{V}} \frac{\sum_{y,x} w_y \left\| \widehat{I}_v(y,x) - I_v^{\text{ref}}(y,x) \right\|_2^2}{3W \sum_y w_y}, \quad (3)$$

where $\mathcal{V}$ denotes the set of optimization viewpoints, $\widehat{I}_v$ and $I_v^{\text{ref}}$ are the panoramic renderings of the current quantized scene and the source scene, respectively, and $W$ and $H$ denote the image width and height. The spherical weight is defined as $w_y = \cos\varphi_y$, where $\varphi_y = \pi((y + 1/2)/H - 1/2)$, such that pixel errors contribute to the distortion according to their corresponding spherical areas.

We then compute the gradient of the spherical distortion with respect to the SH band coefficients of each Gaussian through backpropagation:

$$\mathbf{g}_{i,b} = \frac{\partial D_{\text{sph}}}{\partial \widehat{\mathbf{c}}_{i,b}}. \quad (4)$$

For inactive SH bands, the corresponding coefficients are set to zero while preserving their gradient paths, allowing us to estimate the potential distortion change caused by reactivating these bands. Let $\mathbf{c}_{i,b}^q$ denote the quantized and reconstructed source SH coefficients of band $b$, and let $r_i(k)$ represent the rate estimated by the current conditional entropy model for the attributes and degree signaling of Gaussian $i$ at degree $k$. Adding band $b$ to receiver $i$ induces $\Delta\mathbf{c}_{i,b}^{+} = \mathbf{c}_{i,b}^q$ and

$\Delta\widehat{R}^{+}_{i,b} = r_i(b) - r_i(b-1)$, whereas removing it from donor $j$ induces $\Delta\mathbf{c}^{-}_{j,b} = -\mathbf{c}^{q}_{j,b}$ and $\Delta\widehat{R}^{-}_{j,b} = r_j(b-1) - r_j(b)$. To minimize the rate–distortion cost of SH reallocation, we define the objective as $\min \Delta\mathcal{L}$, with $\Delta\mathcal{L} = \Delta D_{\mathrm{sph}} + \lambda_t \Delta\widehat{R}$, where $\Delta D_{\mathrm{sph}}$ and $\Delta\widehat{R}$ denote the spherical distortion change and the estimated rate change, respectively. The minimization is over candidate exchanges satisfying the SH degree constraints. Since evaluating the actual distortion change for every candidate is computationally expensive, we adopt the first-order approximation $\Delta D_{\mathrm{sph}} \approx \mathbf{g}^{\mathsf{T}}\Delta\mathbf{c}$. The corresponding rate–distortion scores are given by

$$\begin{aligned} s^{+}_{i,b} &= \mathbf{g}^{\mathsf{T}}_{i,b}\mathbf{c}^{q}_{i,b} + \lambda_t[r_i(b) - r_i(b-1)] \\ &= \mathbf{g}^{\mathsf{T}}_{i,b}\Delta\mathbf{c}^{+}_{i,b} + \lambda_t\Delta\widehat{R}^{+}_{i,b}, \\ s^{-}_{j,b} &= -\mathbf{g}^{\mathsf{T}}_{j,b}\mathbf{c}^{q}_{j,b} + \lambda_t[r_j(b-1) - r_j(b)] \\ &= \mathbf{g}^{\mathsf{T}}_{j,b}\Delta\mathbf{c}^{-}_{j,b} + \lambda_t\Delta\widehat{R}^{-}_{j,b}, \end{aligned} \tag{5}$$

where

$$\lambda_t = \frac{D_{\mathrm{sph}}}{\max(\widehat{R}_t, 1)}, \qquad \widehat{R}_t = \sum_i r_i(\ell_i). \tag{6}$$

Combining the receiver and donor changes, the score of an SH capacity reallocation between two Gaussians is expressed as

$$s(i,j,b) = s^{+}_{i,b} + s^{-}_{j,b} = \mathbf{g}^{\mathsf{T}}\Delta\mathbf{c} + \lambda_t\Delta\widehat{R}, \tag{7}$$

where $\mathbf{g} = [\mathbf{g}^{\mathsf{T}}_{i,b}, \mathbf{g}^{\mathsf{T}}_{j,b}]^{\mathsf{T}}$, $\Delta\mathbf{c} = [(\Delta\mathbf{c}^{+}_{i,b})^{\mathsf{T}}, (\Delta\mathbf{c}^{-}_{j,b})^{\mathsf{T}}]^{\mathsf{T}}$, and $\Delta\widehat{R} = \Delta\widehat{R}^{+}_{i,b} + \Delta\widehat{R}^{-}_{j,b}$. Thus, $\Delta\mathbf{c}$ and $\Delta\widehat{R}$ represent the combined coefficient and estimated rate changes of the candidate reallocation. This score combines a first-order distortion approximation with the estimated conditional rate change; a lower score indicates a more favorable predicted rate–distortion change.

*Candidate SH Capacity Reallocation.* For each SH band $b \in \{1,2,3\}$, we construct separate sets of eligible receivers and donors subject to SH degree constraints. Specifically, a receiver $i$ must have a current degree of $\ell_i = b-1$, with its source Gaussian supporting band $b$, whereas a donor $j$ must have a current degree of $\ell_j = b$.

Receivers and donors are independently sorted in ascending order according to their respective scores, $s^{+}_{i,b}$ and $s^{-}_{j,b}$. At most $K$ candidates are retained from each set and paired according to their ranking positions, yielding up to $K$ candidate pairs per band. Only pairs satisfying $s(i,j,b) < 0$ are selected for provisional SH degree reallocation. This rank-based pairing strategy approximately minimizes the predicted rate–distortion cost without exhaustively evaluating all feasible donor–receiver combinations:

$$\ell_i : b-1 \to b, \qquad \ell_j : b \to b-1. \tag{8}$$

The receiver gains one SH band, while the donor discards the same band. SH bands are processed sequentially in ascending degree order, and each Gaussian is allowed to participate in at most one reallocation per iteration. This preserves the SH prefix constraint, the Gaussian count at each SH degree, and the total number of retained SH coefficients. All selected candidate pairs are then combined into a single SH reallocation proposal for joint validation.

*Closed-Loop Rendering and Encoding Validation.* The selected reallocation proposal is provisionally applied to the current Gaussians, which are then re-rendered from the same optimization viewpoints to obtain the updated spherical distortion $D_{\mathrm{new}}$. Only if $D_{\mathrm{new}} < D_{\mathrm{old}}$ do we further adapt the conditional entropy model, perform complete encoding, and verify independent decoding.

The selected reallocation is validated using actual ERP rendering distortion and bitstream size. It is accepted only when

$$D_{\mathrm{new}} < D_{\mathrm{old}}, \qquad B_{\mathrm{actual}} \le B_{\max}. \tag{9}$$

and independent decoding is successfully verified.

Otherwise, the provisional update is reverted, restoring the SH degree configuration from the last accepted iteration. If a proposal exceeds the bitrate budget, alternative candidates may be explored within the remaining iterations.

At the beginning of each iteration, the spherical distortion gradients, conditional rate estimates, and candidate scores are recomputed. The optimization is limited to two iterations and terminates early if no negatively scored candidate pairs remain or if re-rendering fails to further reduce the spherical distortion. All SH capacity reallocation and closed-loop validation operations are performed during offline encoding, introducing no additional optimization overhead during independent decoding.

## III. Experiments

### A. Experimental Setup

We evaluate our method on 43 scenes from four panoramic datasets: 360Roam, OmniPhotos, OmniScenes, and EgoNeRF-Ricoh360, comprising 11, 11, 9, and 12 scenes, respectively [17]–[20]. Experiments are conducted on an NVIDIA RTX 3090 GPU, with native equirectangular projection (ERP) rendering at $1920 \times 960$. Our encoder uses 20-bit geometry quantization and four base attribute bit-width settings, $q \in \{8, 9, 10, 12\}$, corresponding to operating points Q1–Q4. We set the candidate limit to $K = 2048$ for each SH band.

We compare our method with LightGaussian, MesonGS, HAC++, and PCGS [4]–[7], including the SS0–SS2 configurations of PCGS. Reconstruction quality is evaluated against real images at held-out test viewpoints using WS-PSNR [16], LPIPS, V-PSNR, and WS-SSIM. Compression cost is measured by actual storage; for our method, this includes all bytes required for independent decoding. Quality metrics are first averaged over the test views of each scene and then equally averaged across all 43 scenes.

### B. Performance Evaluation

*1) Rate–Distortion Performance:* Fig. 4 compares the rate–distortion performance of the proposed method with HAC++, LightGaussian, MesonGS, and PCGS across 43 panoramic scenes. As the storage cost increases, our method consistently




(a) Ground truth (b) HAC++ (c) LightGS (d) MesonGS (e) PCGS our

Fig. 3. Qualitative comparison on four panoramic scenes. Columns show ground truth, HAC++, LightGaussian (LightGS), MesonGS, PCGS, and our method. Red and blue boxes mark the regions enlarged beneath each panorama. WS-PSNR is reported for each displayed image.

improves across all four quality metrics. Compared with HAC++ and LightGaussian, our method achieves higher reconstruction quality with lower storage at the evaluated operating points. Although MesonGS and PCGS achieve smaller storage sizes at certain operating points, their reconstruction quality is substantially lower than that of our method. These results demonstrate the favorable rate–distortion performance of the proposed panoramic compression framework.

*2) Visual Comparison:* Fig. 3 presents qualitative comparisons of panoramic reconstructions on four representative scenes, where the red and blue boxes indicate regions selected for enlarged visualization. Compared with HAC++, LightGaussian, MesonGS, and PCGS, our method better preserves fine textures and structural details, particularly in regions containing rocks, curtains, and foliage, while reducing visible blurring and reconstruction artifacts. These visual improvements are consistent with the higher WS-PSNR values observed in the corresponding scenes, further demonstrating the effectiveness of our method in panoramic scene reconstruction.

*3) Computational Complexity:* Table I compares the encoding time, decoding time, maximum GPU memory consumption, and ERP rendering speed of different compression methods.

At $q = 8$, our method requires 194.65 s for encoding, longer than the competing methods, as its encoding pipeline involves additional offline spherical rendering feedback and closed-loop validation. The independent decoding time is 18.08 s, slightly shorter than HAC++ (19.57 s) and PCGS (20.74 s), but longer than LightGaussian and MesonGS. In terms of GPU memory consumption, our method requires a maximum of 13.71 GiB, lower than HAC++ (16.73 GiB) and PCGS (20.53 GiB), but slightly higher than LightGaussian and MesonGS.

Furthermore, our method achieves a native ERP rendering speed of 100.548 FPS, approximately $2.41\times$ that of MesonGS (41.721 FPS), the fastest competing method. These results demonstrate that the proposed method enables more efficient native panoramic rendering with moderate GPU memory consumption.

TABLE I
COMPUTATIONAL COMPLEXITY AND ERP RENDERING PERFORMANCE

| Method | Enc. time (s) ↓ | Dec. time (s) ↓ | Max. GPU (GiB) ↓ | ERP FPS ↑ |
|---|---|---|---|---|
| HAC++ | 16.90 | 19.57 | 16.73 | 27.912 |
| LightGaussian | 101.90 | 0.74 | 12.94 | 10.202 |
| MesonGS | 25.48 | **0.563** | **12.31** | 41.721 |
| PCGS (SS1) | **11.19** | 20.74 | 20.53 | 33.196 |
| **Ours** ($q = 8$) | 194.65 | 18.08 | 13.71 | **100.548** |

## C. Ablation of Allocation Components

Table II compares the component configurations of Energy, Pixel, and Spherical. All three use energy-based SH degree initialization. Pixel adds SH capacity reallocation using ERP rendering residuals and gradients, while Spherical additionally

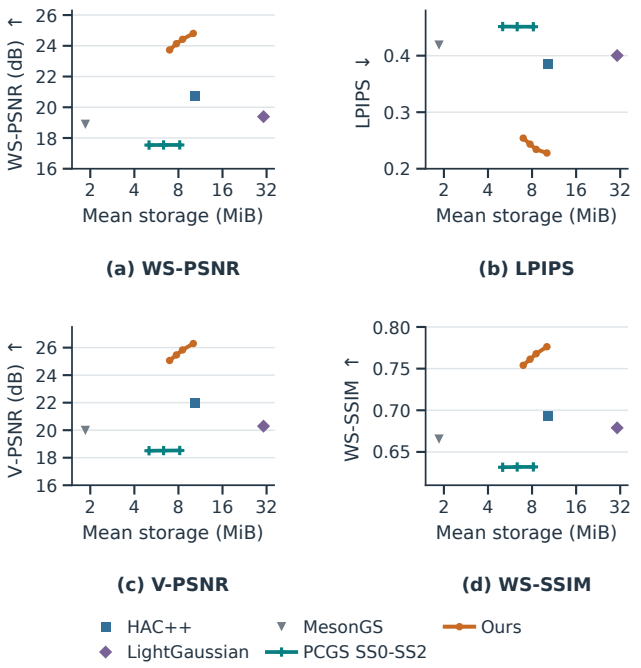



Fig. 4. System-level quality–storage operating points averaged over 43 scenes: (a) WS-PSNR, (b) LPIPS, (c) V-PSNR, and (d) WS-SSIM versus mean storage. Horizontal axes are logarithmic. Lines connect the measured operating points of our method and PCGS SS0–SS2 as visual guides; other external methods are shown as individual measured points. Representation and file accounting conventions are not fully aligned.

incorporates spherical area weighting into the rendering feedback. All reductions in the table use Energy as the baseline and are evaluated over the shared actual byte interval.

TABLE II
ABLATION STUDY OF SH DEGREE REALLOCATION

| Energy Init. | Rendering Feedback | Spherical Weighting | WS-MSE Reduction (%) ↑ |
|---|---|---|---|
| ✓ | × | × | 0.000 |
| ✓ | ✓ | × | 6.533 |
| ✓ | ✓ | ✓ | 7.793 |

Relative to Energy, Pixel and Spherical reduce mean WS-MSE by 6.533% and 7.793%, respectively, improving all 43 scenes. Under the same feedback mechanism, Spherical further reduces WS-MSE by 1.356% relative to Pixel, improving 41 of 43 scenes. Spherical area weighting thus provides additional gains beyond rendering feedback. The latter reduction uses Pixel as its baseline, whereas Table II uses Energy throughout; these percentages therefore cannot be added directly.

## IV. CONCLUSION

We present a panoramic Gaussian compression framework. Residual updates, conditional rate scoring, reversible SH exchanges, and complete byte validation align limited appearance capacity with its rendering value under spherical observations. Full feedback reduces mean test WS-MSE by 7.793% relative to static Energy allocation; spherical weighting provides a further 1.356% reduction relative to Pixel under the same feedback mechanism. The former measures the combined method's benefit, while the latter isolates area weighting. Feedback runs offline, requiring no repeated search at deployment.

## REFERENCES


[1] B. Kerbl, G. Kopanas, T. Leimkühler, G. Drettakis. "3D Gaussian Splatting for Real-Time Radiance Field Rendering," *ACM TOG*, 2023.

[2] L. Diels, M. Vlaminck, W. Philips, and H. Luong, "Fast 3D Gaussian Splatting Rendering via Easily Integrable Improvements," *IEEE Signal Process. Lett.*, vol. 32, pp. 381–385, 2025.

[3] L. Li, H. Huang, S.-K. Yeung, H. Cheng. "OmniGS: Fast Radiance Field Reconstruction using Omnidirectional Gaussian Splatting," *WACV*, 2025.

[4] Z. Fan, K. Wang, K. Wen, Z. Zhu, D. Xu, Z. Wang. "LightGaussian: Unbounded 3D Gaussian Compression with 15x Reduction and 200+ FPS," *NeurIPS*, 2024.

[5] S. Xie, W. Zhang, C. Tang, Y. Bai, R. Lu, S. Ge, Z. Wang. "MesonGS: Post-training Compression of 3D Gaussians via Efficient Attribute Transformation," *ECCV*, 2024.

[6] Y. Chen, Q. Wu, W. Lin, M. Harandi, J. Cai. "HAC++: Towards 100X Compression of 3D Gaussian Splatting," *IEEE Trans. Pattern Anal. Mach. Intell.*, 2025.

[7] Y. Chen, M. Li, Q. Wu, W. Lin, M. Harandi, J. Cai. "PCGS: Progressive Compression of 3D Gaussian Splatting," *AAAI*, 2026.

[8] H. Wang, H. Zhu, T. He, R. Feng, J. Deng, J. Bian, Z. Chen. "End-to-End Rate-Distortion Optimized 3D Gaussian Representation," *ECCV*, 2024.

[9] D. R. Freitas, I. Tabus, and C. Guillemot, "Visibility-Based Geometry Pruning of Neural Plenoptic Scene Representations," *IEEE Trans. Multimedia*, vol. 28, pp. 1–16, 2026.

[10] H. Xu, X. Wu, and X. Zhang, "Improving 3D Gaussian Splatting Compression by Scene-Adaptive Lattice Vector Quantization," *IEEE Trans. Image Process.*, vol. 35, pp. 5864–5879, 2026.

[11] K. Wang, W. Cheng, R. Wang, G. Li, J. Hou, and W. Gao, "PCCRender: Joint Learning of Point Cloud Compression and Gaussian Splatting Rendering," *IEEE Trans. Image Process.*, vol. 35, pp. 8528–8542, 2026.

[12] H. Huang, Q. Yang, M. Liu, W. Huang, Y. Xu, and Z. Li, "Anchor-Driven Compact Gaussian Splatting for Dynamic Scene Reconstruction," *IEEE Trans. Vis. Comput. Graph.*, vol. 32, no. 7, pp. 7188–7201, 2026.

[13] K. Park and C. E. Rhee, "CC-4DGS: Computational Deformation and Point-Cloud Compression for Storage-Efficient Dynamic Gaussian Splatting," *IEEE Trans. Vis. Comput. Graph.*, Early Access, 2026.

[14] Y. Gao, X. Zhang, and G. Zhang, "2D Gaussian Splatting With Pre-Trained Dictionaries for Image Compression," *IEEE Signal Process. Lett.*, vol. 32, pp. 3964–3968, 2025.

[15] Z. Zhao, X. He, S. Xiong, L. He, and R. Sheriff, "An Optimized Rate Control Algorithm in Versatile Video Coding for 360° Videos," *IEEE Signal Process. Lett.*, vol. 29, pp. 2303–2307, 2022.

[16] Y. Sun, A. Lu, and L. Yu, "Weighted-to-Spherically-Uniform Quality Evaluation for Omnidirectional Video," *IEEE Signal Process. Lett.*, vol. 24, no. 9, pp. 1408–1412, 2017.

[17] H. Huang, Y. Chen, T. Zhang, S.-K. Yeung. "360Roam: Real-Time Indoor Roaming Using Geometry-Aware 360-Degree Radiance Fields," *arXiv:2208.02705*, 2022.

[18] T. Bertel, M. Yuan, R. Lindroos, C. Richardt. "OmniPhotos: Casual 360-Degree VR Photography," *ACM TOG*, 2020.

[19] J. Kim, C. Choi, H. Jang, Y. M. Kim. "PICCOLO: Point Cloud-Centric Omnidirectional Localization," *ICCV*, 2021.

[20] C. Choi, S. M. Kim, Y. M. Kim. "Balanced Spherical Grid for Egocentric View Synthesis," *CVPR*, 2023.